

Code Purpose

This repository contains a collection of sample code and source code in MATLAB for aggregating data from Snapshot and Log folders' converted CSV files using Kitronyx products.

FolderTree

```
| README.md  
| README.pdf  
├─res  
├─sample  
|   ├──calculate_drift_rate  
|   |   └─converted_log_data  
|   ├──data_statistics  
|   |   └─snapshot_data  
|   └─read_kitronyx_csv  
|       ├──converted_log_data  
|       └─snapshot_data  
└─src  
    ├──calculate_drift_rate  
    ├──data_statistics  
    └─read_kitronyx_csv
```

MATLAB

Version: R2023b Update 7 (23.2.0.2515942) 64bit January 30, 2024

How to Use Sample Code

To use the sample code in each folder:

1. Navigate to the desired sample folder (e.g., sample/calculate_drift_rate, sample/data_statistics, sample/read_kitronyx_csv).
2. Run the '**main.m**' file in the folder to execute the corresponding sample code.

This will allow you to test and observe how each function processes the data.

Code Description

```
read_snapshot_1d_data.m  
- MATLAB file containing a function to read snapshot 1D files  
- Returns [row, col, data] when given a 1D CSV path as a parameter.  
- row: ROW - number of columns
```

- col: COL - number of rows
- data: Cell array data (size ROW*COL)

read_converted_logfile_1D_data.m

- MATLAB file containing a function to read log 1D files
- Returns [row, col, times, data] when given a 1D CSV path as a parameter.
- row: ROW - number of columns
- col: COL - number of rows
- times: Cell array - Time values
- data_dict: Cell array data (size ROW*COL)



calc_node_sum_max_min_avg.m

- MATLAB file containing a function to calculate sum, average, maximum, and minimum values for all nodes.
- Returns [nodeSum, nodeMax,nodeMin,nodeAvg] when given a 1D matrix data as a parameter.
- nodeSum: Sum of all nodes
- nodeAvg: Average of all nodes
- nodeMax: Max value of all nodes
- nodeMin: Min value of all nodes

clac_node_rsd.m

- MATLAB file containing a function to calculate Standard deviation and Relative Standard deviation values for all nodes.
- Returns [nodeStd, nodeRsd] when given a 1D matrix data as a parameter.
- nodeStd: Standard deviation of all nodes
- nodeRsd: %RSD of all nodes

calc_node_xrad.m

- MATLAB file containing a function to calculate %XRAD values for all nodes.
- Returns [nodeXrad] when given a 1D matrix data as a parameter.
- nodeXrad: %XRAD of all nodes

```

main.m
명령 창
sum of all nodes:
    483883

max value of all nodes:
    211

min value of all nodes:
    209

average of all nodes:
    210.0187

Standard deviation of all nodes:
    0.2502

%RSD of all nodes:
    0.1191

%XRAD of all nodes:
    0.4673

>>

workspace
이름      값      클래스
col      48      double
data     1x2304 do... double
node_avg 210.0187 double
node_max 211      double
node_min 209      double
node_rsd 0.1191   double
node_std 0.2502   double
node_sum 483883   double
node_XRAD 0.4673   double
row      48      double

```

calcuale_drift_rate.m

- MATLAB file containing a function to calculate drift rate for all log data
- Returns [drift_rate, driftInfo] when given a drift value and drift information struct.
- drift infomation struct format example:
 - driftInfo.timeStart
 - driftInfo.timeEnd
 - driftInfo.adcBegin
 - driftInfo.adcEnd

```

Drift start time: 1, Drift end time: 7488.03, Drift start ADC: 76, Drift end ADC: 80
Drift rate of all log: 1.3585 %log10time
>>

driftInfo
1x1 struct with 4 fields
Field      Value
timeStart  1
timeEnd    7.4880e+03
adcBegin   76
adcEnd     80

workspace
Name      Value
driftInfo 1x1 struct
driftRate 1.3585
measuredCol 66

```